Toward Clinical Dark-field CT with Physics-guided Neural Field Reconstruction

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Dark-field computed tomography (DFCT) is an emerging X-ray imaging modality based on small-angle scattering that provides functional information for the improved diagnosis of pulmonary diseases [1]. Recently, a human-scale prototype has been realized by adopting gratings into a commercial clinical gantry [2]. Despite advanced algorithms capable of recovering phase information from the continuously rotating system, streak artifacts remain a crucial challenge for clinical translation [3]. The setup imposes hard constraints on engineering-based improvements, and classical data processing faces inherent limitations due to the ill-posed nature of X-ray phase retrieval. This motivates the exploration of learning-based image enhancement strategies.

To this end, we investigate continuous neural representations for inverse problem solving. Neural field reconstruction is constrained to physically consistent outcomes by forward-mapping projections via a Talbot–Lau interferometer model to measurement space and penalizing deviations from the recorded signal. Thereby, we decouple phase retrieval from tomographic reconstruction, allowing for independent optimization at both stages. This strategy reduces the risk of introducing spurious structures into the reconstruction, as hallucinated features in projection space are unlikely to occur consistently across multiple angles to accumulate coherently. The self-supervised, per-instance formulation enables phase-refined reconstruction without requiring a dedicated ground truth and circumvents potential domain gaps arising from a lack of clinical training data.

Our method improves peak signal-to-noise ratio (PSNR) from 34.8 dB to 38.5 dB (+3.7 dB) and structural similarity index measure (SSIM) from 0.831 to 0.938 (+0.107), evaluated for a reconstructed volume against a pseudo-ground-truth at 10-fold dose obtained from consecutive gantry rotations. These findings highlight that integrating physical domain knowledge with learned representations constitutes a step toward robust DFCT phase retrieval and the eventual translation from a prototype system into clinical practice.

[1] Willer et al., Lancet Dig Health, 2021
[2] Viermetz et al., PNAS, 2022
[3] Haeusele et al., IEEE TMI, 2023